

Hidden Parameters and Learning Deadlines: Structural Uncertainty as a Timing Obstruction

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Abstract

Many sustainability failures are caused not by noisy data but by structural uncertainty resolved too late. This paper, the D10 item of the generalization catalogue, formalizes that sentence inside the obstruction calculus. The state augments with a hidden parameter $\theta \in \Theta$; the belief lives on the product; the obstruction is the emptiness of the joint floor-servable set over (x, θ) cells. On the declining working instance ($\theta = \pm 1$ gating the correctable component of a $-\frac{1}{10}$ drift), the theory computes three exact thresholds: with θ observed, every grid cell is viable; under one-probe endogenous learning the probe reveals θ exactly (the branch separation is 2, so one observation splits the belief with no statistics) but survival forces $z_0 \geq \frac{21}{10}$; and with exogenous revelation at horizon T_{learn} viability holds exactly iff $z_0 \geq \frac{21}{10} + \frac{T_{\text{learn}}}{10}$ — the learning deadline acts as the timing obstruction’s observation deadline, and the kernel shrinks exactly linearly in it ($|K| = 5, 4, 3, 2$ on the grid for $T_{\text{learn}} = 0, 1, 2, 3$), antitone with strict steps. A two-patch instance with unknown regeneration rate $g \in \{1, 2\}$ shows the same phenomenon on the programme’s audited belief: unknown-regeneration nonviable, (x, g) -observed viable (6/6 check families).

Introduction and relation to the programme

Paper 2’s hidden-mode example ($\dot{z} = \theta u$, $\theta = \pm 1$) is the seed; the catalogue generalizes it to augmented states $(x, \theta) \in X \times \Theta$, beliefs on the product, and the learning deadline T_{learn} in the role of the timing layer’s observation deadline, with endogenous learning as belief updating. The stochastic selector supplies the belief-dynamics discipline; the successor supplies the timing object; neither is modified. The list of affected sustainability parameters — regeneration rates, climate sensitivity, institutional effectiveness — motivates; the paper verifies on exact instances.

Hidden parameters in the calculus

The augmented belief $B \subseteq X \times \Theta$ is read per cell under the successor’s read-then-act semantics; the obstruction is

$$\bigcap_{(x, \theta) \in B} \mathcal{R}_V(x, \theta) = \emptyset \implies B \notin \mathcal{K}_{\mathcal{I}}.$$

Learning enters as information: exogenous revelation at T_{learn} (the structure coarsens before, refines after) or endogenous probing (actions chosen to split the belief).

Proposition 1 (Learning deadline as timing obstruction). In the declining working instance with quantized control $u \in \{-1, 0, 1\}$ and hold-then-probe-then-act schedules: viability holds exactly iff $z_0 \geq \frac{21}{10} + \frac{T_{\text{learn}}}{10}$. The kernel is antitone in T_{learn} with strict steps, and the probe reveals θ exactly (the two post-probe branches differ by 2, so one observation splits the belief).

Proof. Before T_{learn} the drift is $-\frac{1}{10}$ (hold $u = 0$), so survival to the probe forces $z_0 - \frac{T_{\text{learn}}}{10} \geq$ the probe threshold; the probe $u = +1$ moves z by $-\frac{1}{10} + \theta$, which must stay safe on both branches, giving the threshold $\frac{21}{10}$ at $T_{\text{learn}} = 0$; after the probe the learned law $u = \theta$ has drift $-\frac{1}{10} + \theta^2 = \frac{9}{10} > 0$, so survival is indefinite. The branch positions $z_0 - \frac{1}{10}T_{\text{learn}} - \frac{1}{10}\pm 1$ differ by 2, so the post-probe reading determines θ uniquely. Antitonicity is the threshold identity; strictness on the grid is the computed counts 5, 4, 3, 2. $\square$

Verified record (6/6 exact check families)

E1 θ -observed viability on all 16 grid cells (drift $\frac{9}{10}$); E2 the probe threshold $\frac{21}{10}$ as a branchwise identity; E3 the deadline identity at $T_{\text{learn}} = 0, \dots, 4$ (at 4 exactly the top cell $\frac{25}{10}$ survives); E4 the exact separation of the post-probe branches (injective revelation, no statistics); E5 the antitone chain with exact counts $|K| = 5 > 4 > 3 > 2$; E6 the two-patch unknown-regeneration instance: the audited belief with unknown $g \in \{1, 2\}$ is nonviable under the g -blind aggregate reading and viable under (x, g) -observed cells, each fixed g viability-preserving — structural uncertainty, not noise, is the obstructing object.

What is not claimed

No learning theory is claimed beyond the one-probe exact split: Bayesian updating, repeated experiments, and exploration–exploitation trade-offs are out of scope. The deadline result

is for the audited schedule family (hold, probe, learned act); other schedules are not classified. The parameter θ enters additively; state-dependent parameter dynamics and parameter drift belong to the hybrid companion’s setting. The sustainability parameter list motivates the model class and carries no empirical content here.

Declarations

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Data availability. No external data were used.
Code availability. The verification script `hidden_parameter_learning_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.